

CITYTWIN · DESIGN DECISION RECORD

CityTwin recommendations accepted

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DOCUMENT CONTROL

Document ID	CTW-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/citytwin		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	O1, Q1 and Q2 decided by Amish on 2026-10-02 as recommended
0.1	2026-09-25	Amish Chadha	Record the recommendations accepted by Amish on 2026-09-25, what changed in the repo and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item below that carried a recommendation is decided by Amish, 2026-09-25: go with recommendation. Items with no recommendation stayed "Proposed, awaiting Amish"; O1, Q1 and Q2 were decided on 2026-10-02 as recommended in CTW-DEC-001 (Amish, 2026-10-02: "i approve your recommendations for all 555 open decisions.").

■ Context

On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos." Before that, CTW-DDR-001 (D1 to D11) recorded this repo's TRL 2 review choices as adopted for TRL 3 work and open for his review, and the TRL 3 review note (docs/REVIEW.md , session 2026-09-25: TRL 3) listed six further items with options and a recommendation. This record lists every item now decided, what changed in the repo because of it, and what remains open. Where a recommendation offered several options, the recommended option is the decision. Work that belongs to TRL 4 (build, test, measurement, purchasing, firmware beyond a sketch) is decided but on hold, because TRL 4 is on hold by Amish's instruction.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in the repo
D1	Budget scope (CTW-DDR-001)	Option (b): the budget covers the kiosk; the TwinKit gateway is costed in TwinKit	Status wording only; R17 stays a reported figure
D2	Display	Color e-paper for the street kiosk, indoor LCD variant	Status wording; the indoor LCD variant is named in R10 and remains unmodeled at TRL 3
D3	Climate range	Sheltered or indoor pilot, heater study at TRL 3	Status wording (range narrowed by N5 below)
D4 to D10	Gateway in the cabinet; mains power with a solar variant; 5 per hour threshold and two years of raw data; CC BY 4.0; buttons; 46-node reference neighborhood; one-way publishing with privacy rules at the gateway	As recommended	Status wording in CTW-DDR-001 v0.2, CTW-PRB-001 v0.4, CTW-PRC-001 v0.4 and CTW-REQ-001 v0.4
D11	Pull paths for PotholeLog and DockHub	The gateway pulls published PotholeLog segment files and DockHub hourly aggregates by outbound HTTPS	Status wording; format agreement listed as a cross-repo action in <code>docs/REVIEW.md</code>
N1	Budget figure	<code>budget_usd</code> raised to the recommended \$750	<code>project.yaml</code> <code>budget_usd</code> \$600 to \$750; R16 now met on paper for the pilot kiosk (\$739.00, \$11.00 under); README budget line updated
N2	R10 button response	Option (a) for the street kiosk and (b) indoors: press acknowledged within 0.5 s by the button's LED ring and layer shown within 25 s; the indoor LCD variant keeps 5 s	R10 restated in CTW-REQ-001; firmware rule "LED ring lit from the press until the new image is drawn" added to CTW-PRC-001; CTW-CAL-001 [C2] checks both limits; R10 not met (19.8 s against 5 s) to met on paper (30 ms and 19.8 s against 0.5 s and 25 s)
N3	R18 outage ride-through	Option (b): ask TwinKit for a larger pack in cabinet installs	CTW-CAL-001 [I1] now sizes the pack: at least 1.84 Ah against TwinKit's 1.5 Ah. Request listed as a cross-repo action; TwinKit not edited. R18 stays not met until TwinKit changes its pack
N4	R20 street climate and the cabinet	A ventilated sun shield on the cabinet and a shaded screen orientation as site rules for any outdoor pilot; a panel heater only if the partner city has frost	New BOM item 19, cabinet sun shield (1.5 mm aluminum, 413 x 186 x 486 mm, 25 mm air gap, open bottom, top vents), \$30.00, in <code>cad/src/model.py</code> , STEP and STL, CTW-DWG-001 Rev P2 and all media. CTW-CAL-001 v0.2: cabinet rise in sun 18.7 K without the shield, 7.2 K with it; the pack stays under its 45 °C charge limit in full sun up to 37.8 °C air. Site rules added to CTW-PRC-001. R20 stays not met (panel 45.1 °C shaded at 45 °C air)

#	Item	Decision	What changed in the repo
N5	R12 pilot margin	Define the pilot range as 0 to 35 °C air	R12 target 0 to 40 °C to 0 to 35 °C; panel 35.1 °C (4.9 K margin), pack 39.4 °C (5.6 K margin); R12 at risk to met on paper
N6	Hood light (suggestion in the TRL 3 review note)	Dim the 2 W hood light to about 0.3 W, since 200 lx is enough	Controller rule in CTW-PRC-001; CTW-CAL-001 [D2, G1]: 219 lx at night, mains power 10.2 W to 9.2 W, 90 to 81 kWh a year

Table 2. Items still open.

#	Item	Status
O1	First partner city or neighborhood for co-design and a pilot site. No recommendation was made.	Decided 2026-10-02: a city within easy reach of Irving, Texas, with an active open-data program, the first candidate to approach; sheltered public lobby, no frost heater
Q1	Antenna height needed for the reference neighborhood with the gateway in the cabinet	Decided 2026-10-02: post-top antenna at 2.38 m for the pilot, coverage walk test at TRL 4, a rooftop gateway rather than a taller mast if nodes fall short
Q2	Languages and scripts on the kiosk, and how residents ask questions or report a fault	Decided 2026-10-02: English plus the partner city's most widely spoken other language in its own script, with a QR code and a phone or text number on the notice plate; refined in co-design

■ Consequences

- `project.yaml`: `budget_usd` \$750. Pitch and problem unchanged: no rewording was recommended. `trl`: 3 and `trl_target`: 3 unchanged.
- Documents revised: CTW-PRB-001 v0.4, CTW-PRC-001 v0.4, CTW-REQ-001 v0.4, CTW-CAL-001 v0.2, CTW-DDR-001 v0.2; drawing CTW-DWG-001 Rev P2; `bom/bom.csv` (19 lines) and `bom/bom-notes.md`; `README.md`.
- Requirement status (CTW-CAL-001 v0.2): 2 not met (R18, R20), 1 at risk (R1), 10 met on paper, 7 met by design. Before: 4 not met, 2 at risk, 7 met on paper, 7 met by design.
- The sun shield adds 1.86 kg and 35 N of gust load: post stress 26.2 to 26.9 MPa, anchor tension 1.92 to 1.97 kN, mass 58.2 to 60.1 kg. All still met.
- Cross-repo actions (TwinKit pack, PotholeLog and DockHub pull formats, CrossSafe uplink) are listed in `docs/REVIEW.md`. No other repo was edited.
- On hold at TRL 4 by Amish's instruction: the panel heater (only for a frost site), the indoor LCD and solar variants beyond their mention, and every build, bench timing, climate chamber and outage test. Nothing in this record authorizes building or testing.